

CS3.301: Operating Systems and Networks

Lecture 8 - Introduction to Networks

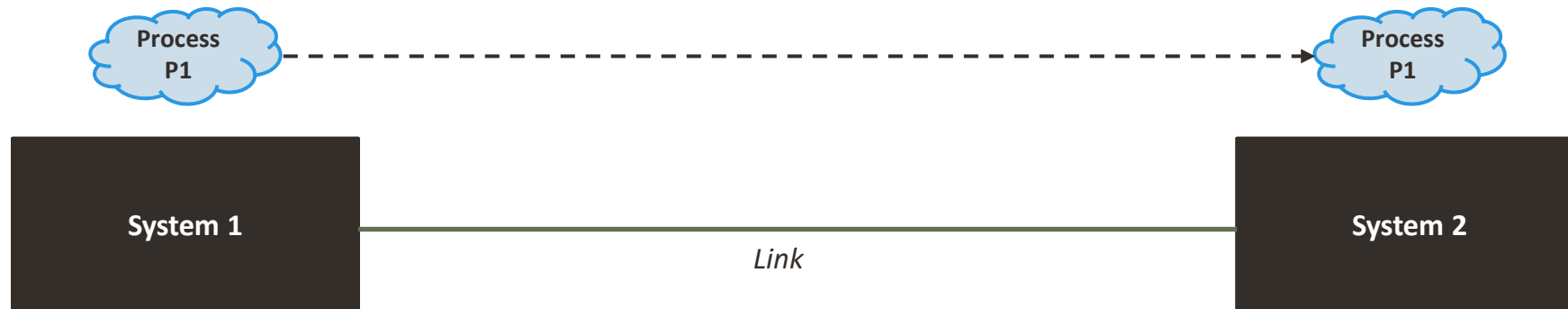
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Acknowledgement

- The material in this lecture has been gathered, adapted, and generated from multiple sources, combined with the instructor's own experience and knowledge.
- Sources:
 - Computer Networks, 6th Edition — Tanenbaum, Feamster, and Wetherall
 - Computer Networking: A Top-Down Approach — Kurose and Ross
 - Computer Networking Essentials — YouTube channel
 - Other online sources, duly cited where used

What About Processes in Different Machines?

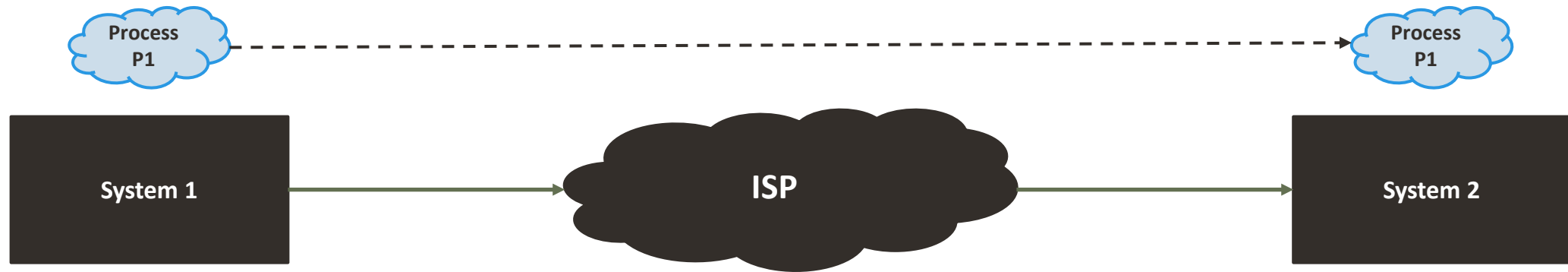
How can they communicate?



How does a message/data from P1 in System 1 reach P1 in System 2?

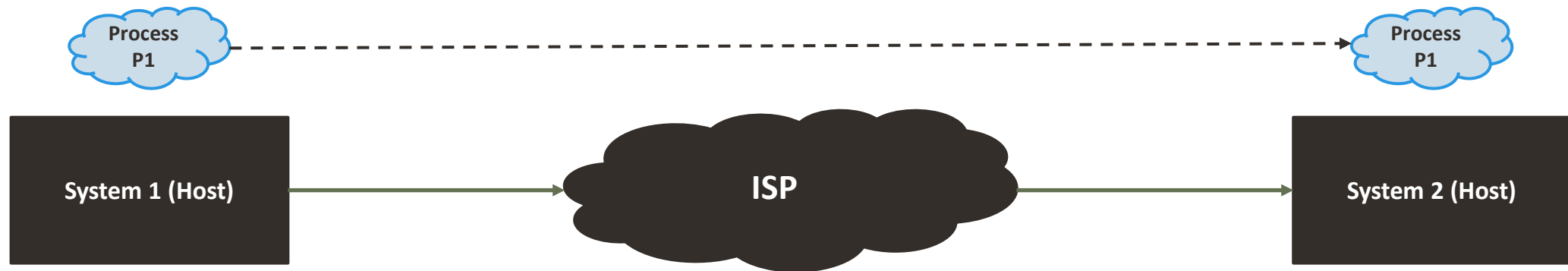
What role does the OS play, and how does it contribute to effectiveness?

Let Us Expand It



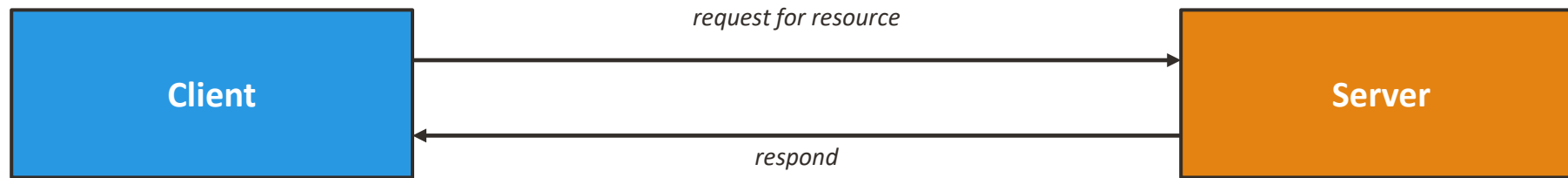
- Between the process and the network, there needs to be an interface.
- Between the network components themselves, there also needs to be some interface.

Network Components: Host

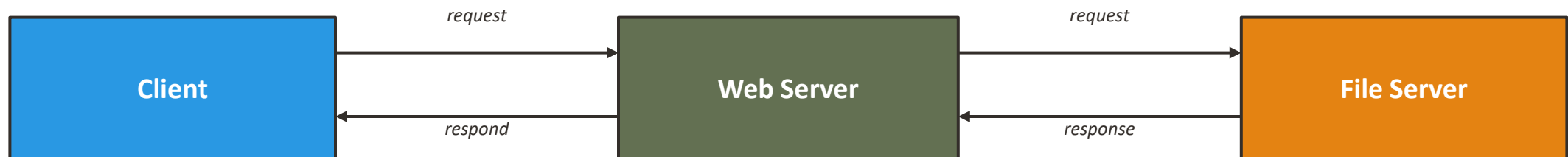


- A host is any device that sends or receives traffic — computer, laptop, smartwatch, phone, etc.
- A host can act as a client or a server.
- A server can sometimes act as a client too, when it makes requests of another server.

Client–Server



A server can also act as a client



How Does a Host Send Data? — IP Address

- A host needs an address to send data — this address is known as its IP address.
- When a client sends data, it provides both a source IP and a destination IP.
- IP addresses are 32 bits long; each bit is 0 or 1.
- They are written as 4 octets, each ranging from 0–255 (8 bits each).
- IP addresses are usually assigned hierarchically.



192.168.1.1

example IP address

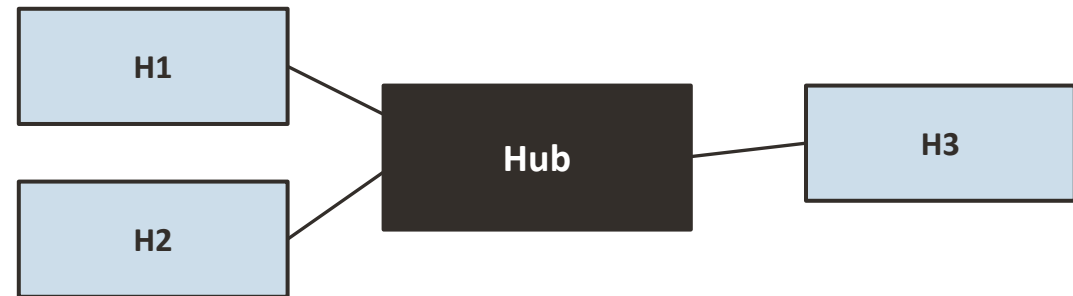
Sometimes We Need More Signal Strength — Repeater

- A repeater allows regeneration of signals for long-distance communication.
- Think of Wi-Fi in a home:
 - Signal strength may not reach every room.
 - A repeater may be needed to transmit to longer ranges.
- But we cannot just connect one host directly to another — that approach is not scalable!

Hubs and Bridges

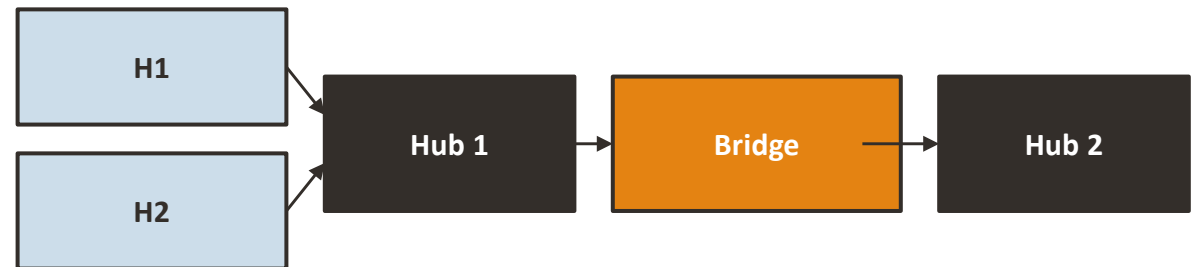
■ Hubs

- Multi-port repeaters.
- Key issue: everyone receives everyone's data.



■ Bridges

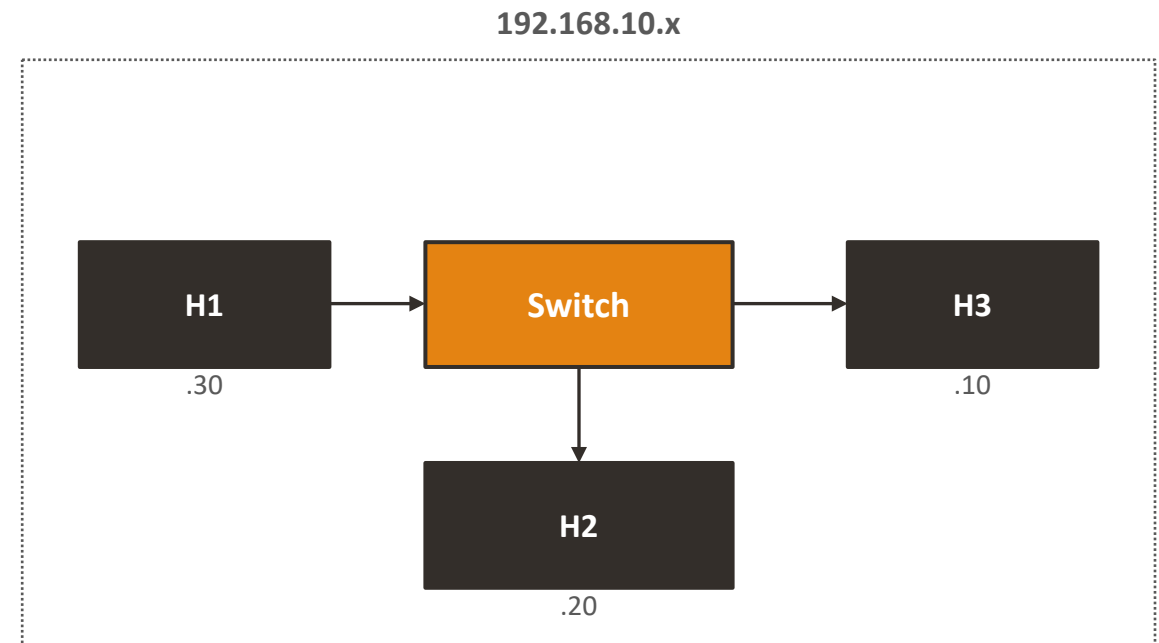
- Sit between two hubs.
- They have only two ports.
- They learn which hosts are on either side (for routing).



e.g., H1 wants to communicate with H2

Switches

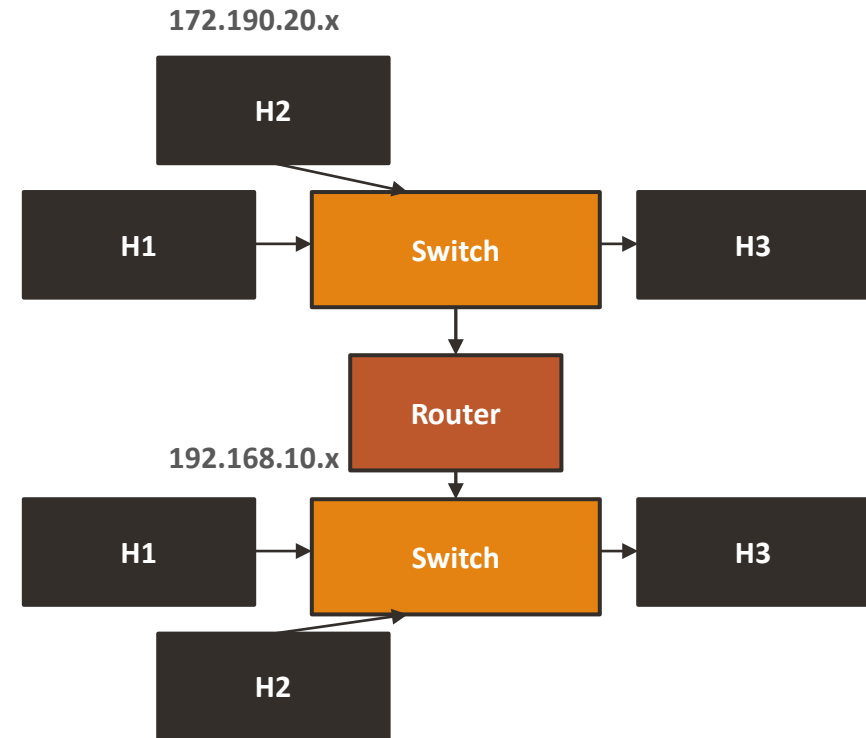
- Devices which facilitate communication within a network.
- A combination of hubs and ports.
- Knows or infers which hosts are on each port.
- They have multiple ports.
- Whatever is connected to a switch becomes part of one network.



What if host 192.168.10.20 wants to talk to a host in a different network?

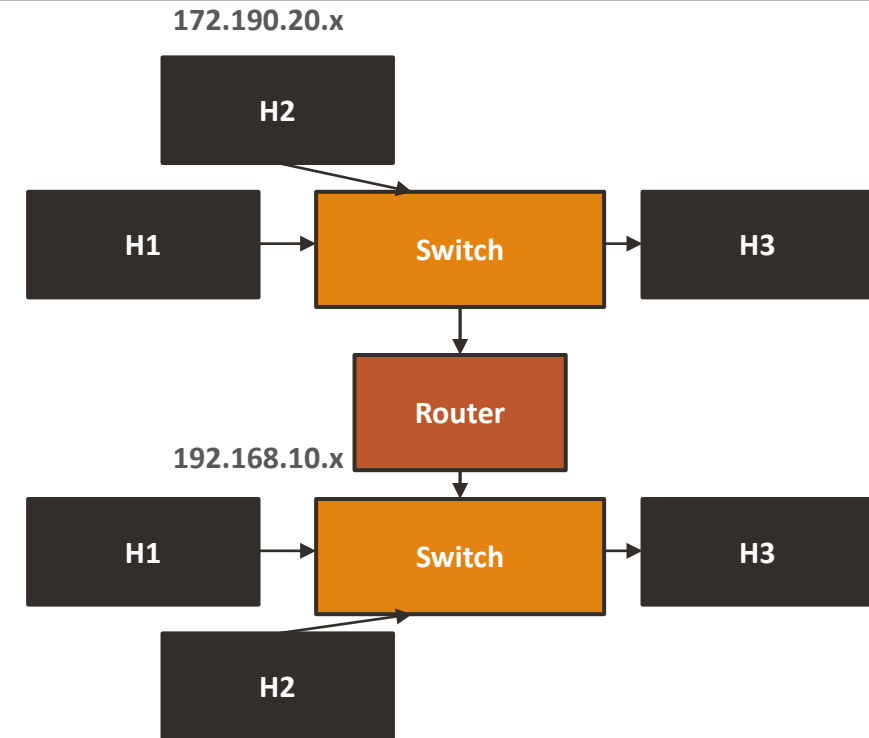
Routers

- Facilitate communication between networks.
- They act like a traffic control point:
 - Security, filtering, redirection.
- Routers learn which networks they are attached to.
 - Known as routes — stored in a routing table.
- Routers have their own IP address in each network they are attached to.

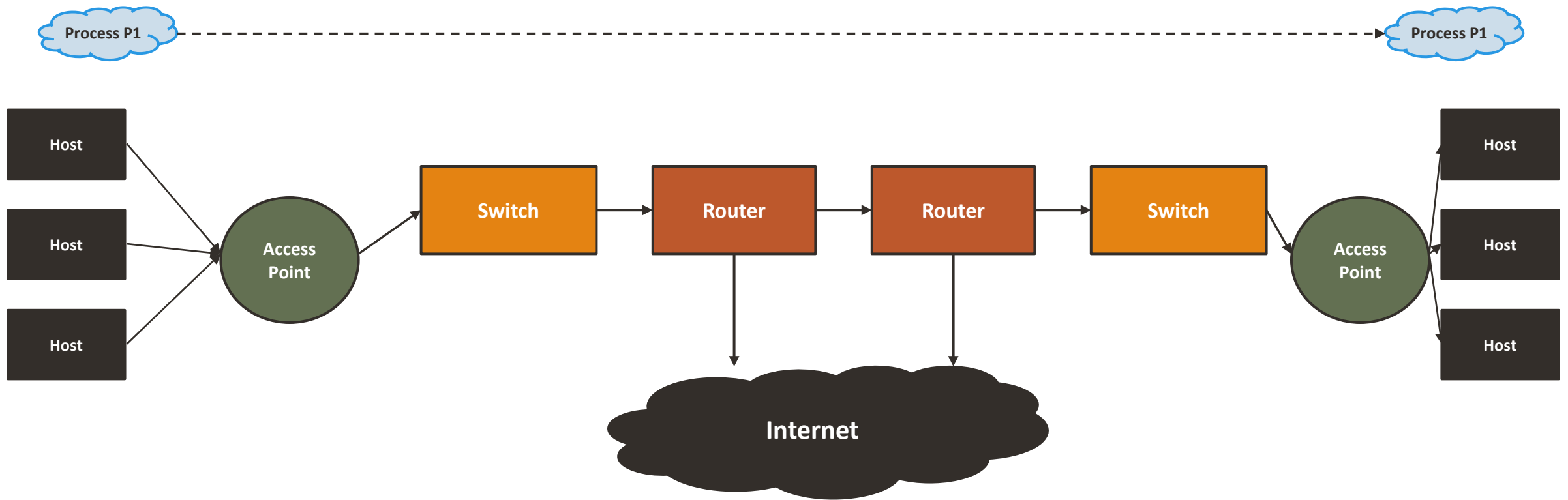


Routers — Building a Hierarchy

- Allow creation of hierarchy in networks.
- Every time a host wants to go outside its own network, traffic goes through the router.
- The Internet is nothing but a huge collection of routers.
- Gateway = the IP address of the router in a given network.
 - Becomes the exit point for hosts leaving that network.



The Bigger Picture



What Happens Internally?

What happens internally when one process wants to share or receive data from another host?

Network Types

Multiple types of networks exist

- Personal Area Network (PAN)

- Devices communicate over a very short range.
- E.g., Bluetooth (master–slave paradigm).

- Local Area Network (LAN)

- A private network operating within/near a single building (home, office, factory, etc.)
- Wireless LANs run from 11 Mbps to 7 Gbps (powers of 10, not 2) — 1 Mbps = 1,000,000 bits/sec.
- Wired LANs operate from 100 Mbps to 40 Gbps, with low latency.
- E.g., connecting personal computers and consumer electronics such as a printer.

Network Types — We Have Come a Long Way!

- One large physical LAN can be divided into smaller logical LANs (Virtual LANs, or VLANs).
- In earlier days, it was all about broadcasting on a single shared line!
 - At most one computer could transmit successfully at a time.
 - Static allocation techniques were used:
 - Every machine got some time to transmit or receive.
 - Round robin was used for scheduling.
 - Packet collisions used to happen (wait a random time and try again).

Network Types

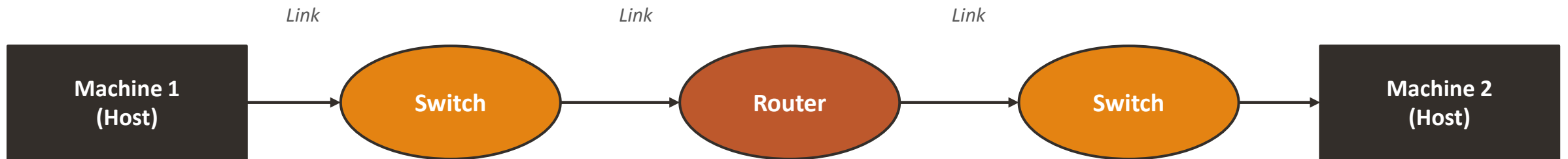
- Metropolitan Area Network (MAN)

- Covers a city (city-wide networks).
- Think of cable TV networks — earlier, an antenna on top of the house, and from the top of a hill, communication used to happen.
- The idea of cable TV was used to provide two services: Internet and TV.

- Wide Area Network (WAN)

- Spans a large geographical area — often a country or continent.
- E.g., the Internet is a large WAN. Dedicated WANs also exist.
- A WAN can connect offices of an organization across different locations.
- Higher latency and lower transmission speeds; dedicated WANs can be costly.

Components of a Network



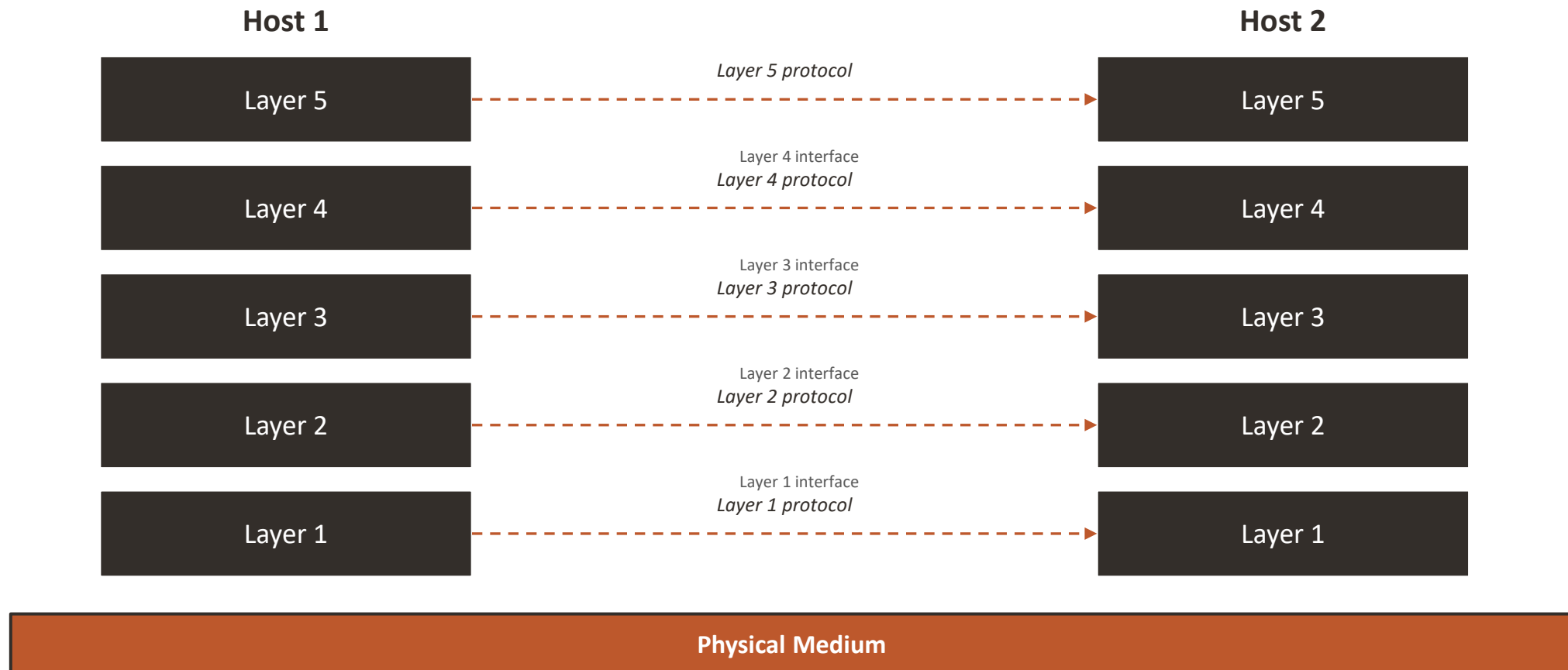
Process Communicating Over a Network

- Process A (e.g., WhatsApp) is executing on Host 1.
- Process B (WhatsApp) is executing on Host 2.
- Process A wants to communicate with Process B.
 - A system call is made to access the network.
 - This leads to an interrupt.
 - Process A should know where Process B is.
 - Hardware support is needed — some network device with drivers.

Let's Take This to Process Communication

- Process A (e.g., WhatsApp) is executing on Host 1.
 - Process B (WhatsApp) is executing on Host 2.
- Host 1 will have an address; the same is true for Host 2.
- How do we ensure the data reaches from Host 1 to Host 2?
 - What all needs to be considered?
- Remember: there will be multiple processes executing on a host at once.

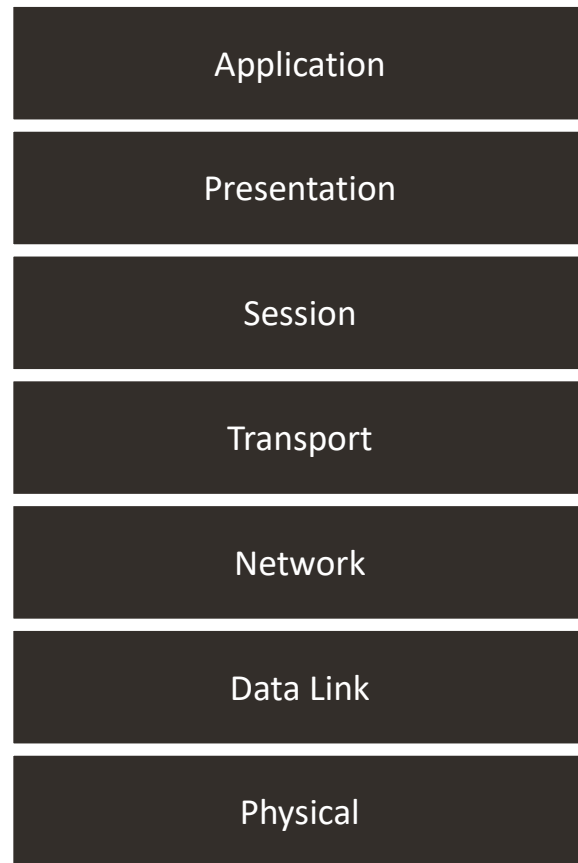
Networking Layers



Networking Layers

- Layering reduces design complexity — a network is organized as a stack of layers, or levels.
- The layering provides abstraction to and from other layers.
- Communication, in some sense, happens between corresponding layers.
 - Layer n on one machine communicates with layer n on another using a protocol.
 - A protocol is just an agreement between communicating parties on practices.
- A set of layers together with their protocols is called a Network Architecture.
- Between each pair of adjacent layers there is an interface.
 - It defines the primitive operations and services the lower layer makes available.

The OSI Model

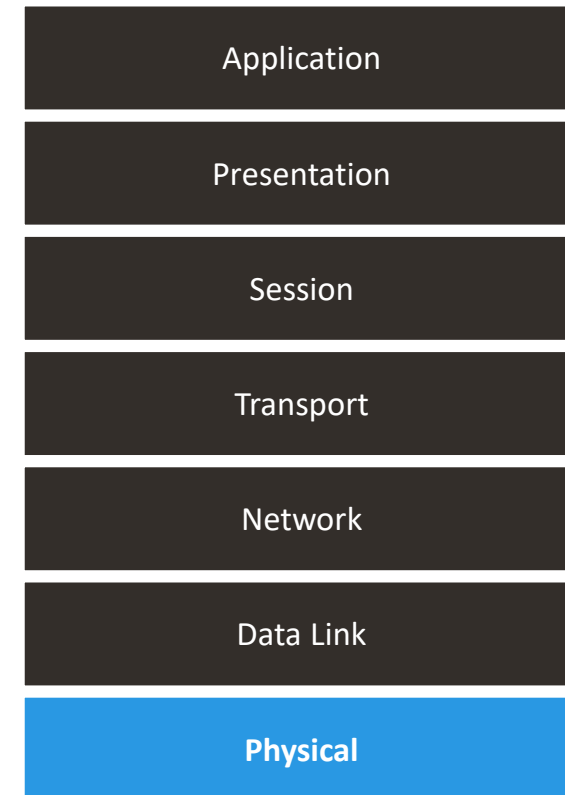


- Open Systems Interconnection (OSI).
- A conceptual framework used to understand how communication works through different layers.
- Divides the network communication process into seven layers.
- Developed to facilitate interoperability between different technologies.
- Each layer has a specific function — if they all do what they're supposed to, the result is successful sharing of data.

Physical Layer (L1)

Ultimately, everything is 0's and 1's

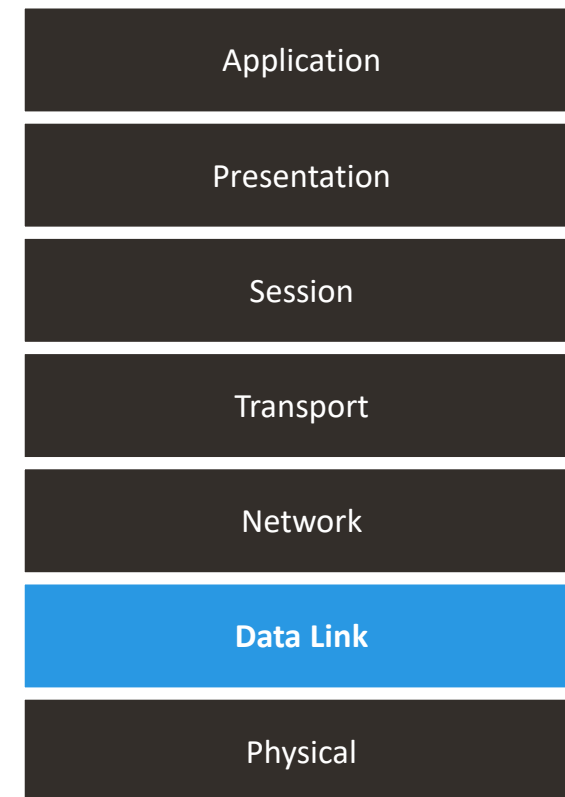
- Data is in the form of bits — 0s and 1s.
- Something has to transport those bits from one machine to another — that's the physical layer.
- Concerned with transmission of raw bits over a physical medium, like a cable.
- L1 technologies: Ethernet cables, optical fiber, coaxial cable, etc.
 - Even Wi-Fi is an L1 technology — as are hubs, repeaters, etc.



Data Link Layer (L2)

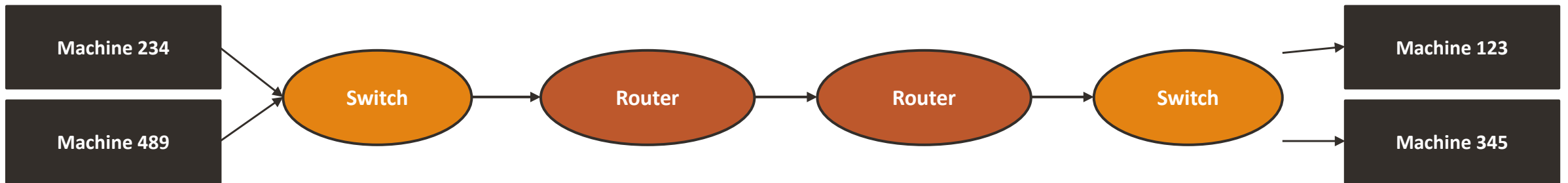
Hop-to-hop connectivity

- Interacts with the physical medium.
- Adds/takes data to and from the L1 technology.
- Responsible for creating a reliable link between two directly connected nodes, error correction, etc.
- L2's responsibility is mainly taking data from one hop to the next.
 - Uses an addressing scheme — MAC addressing (48 bits, written as 12 hex digits).
- L2 technologies: NICs, Wi-Fi access cards. Switches are also an L2 technology.



Communication Is More Than Hop-to-Hop

What about communication from Machine 234 to Machine 345?

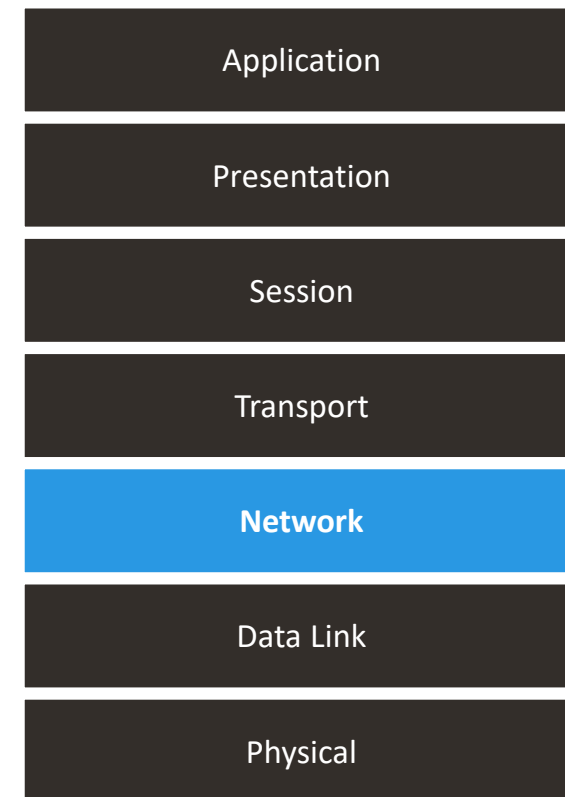


- Hop-to-hop delivery is guaranteed by L2 (Data Link Layer).
- But what about end-to-end delivery across multiple hops?

Network Layer (L3)

End-to-end communication

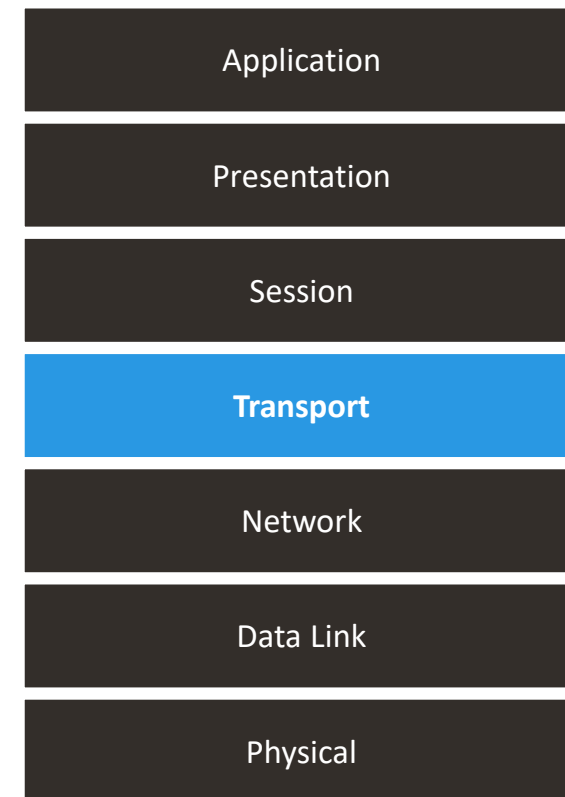
- Manages routing through different routes in a large network.
- Uses an addressing scheme — IP addressing.
 - 32 bits, represented as 4 octets (IPv4).
- Performs functionalities such as logical addressing (IP), path selection, and packet forwarding.
- L3 technologies: routers — even hosts operate at L3 — and L3 switches.



Transport Layer (L4)

Service-to-service

- How do we ensure the right process receives the data?
 - Many processes may be executing, or waiting to execute, on the same host.
- Ensures data transfer is reliable, sequential, and free from errors.
- Manages flow control and error correction.
- Layer 4 has an addressing scheme to guarantee message delivery: ports!
 - Ports range 0–65535: Privileged 0–1023, Registered 1024–49151.
 - Every process sends/receives data through a port.
- L4 technologies: TCP, UDP.



Session, Presentation, and Application

Application-to-application

■ Session Layer (L5)

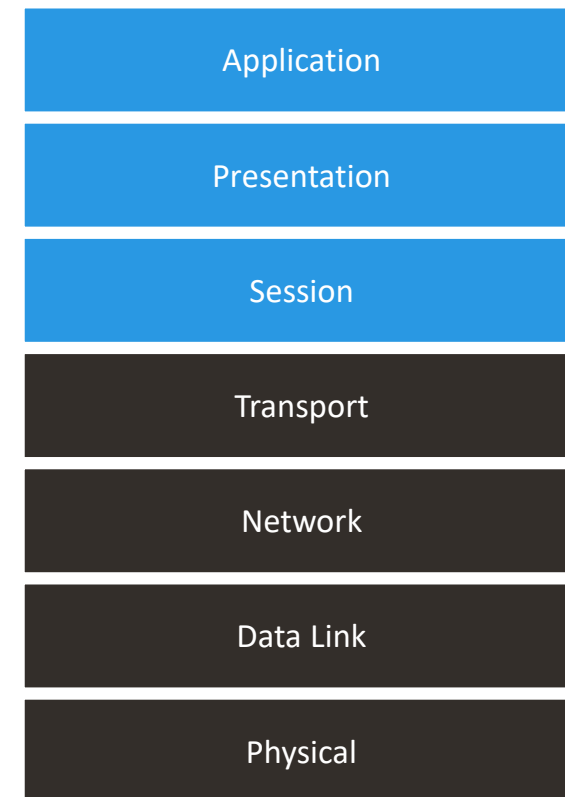
- Manages the connection between different devices.
- Establishing, maintaining, and terminating connections.

■ Presentation Layer (L6)

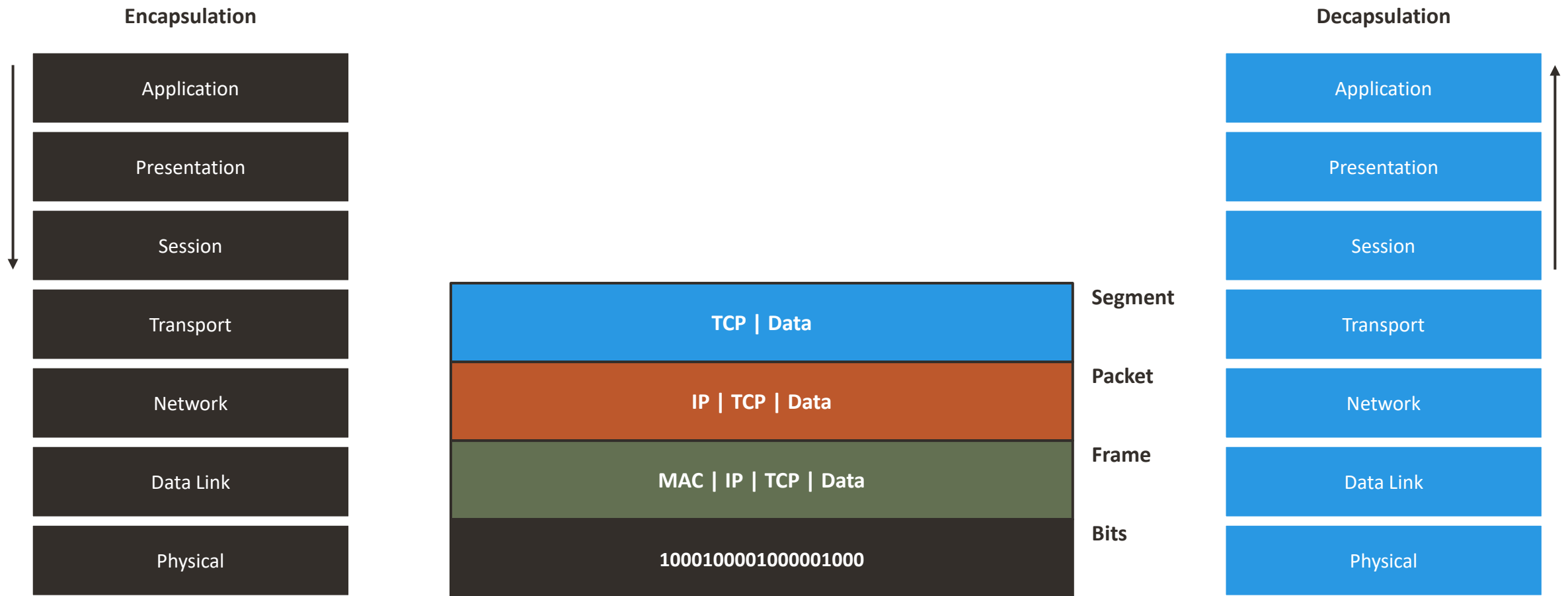
- Ensures data is in a format both sender and receiver can understand.
- Manages data encryption and compression.

■ Application Layer (L7)

- Provides network services to application processes.
- E.g., web browsers, email clients, other software/apps.



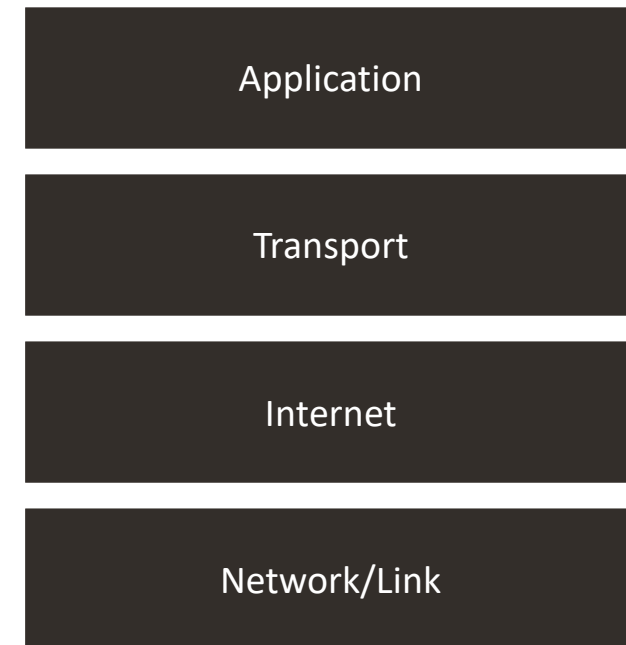
Putting It Together



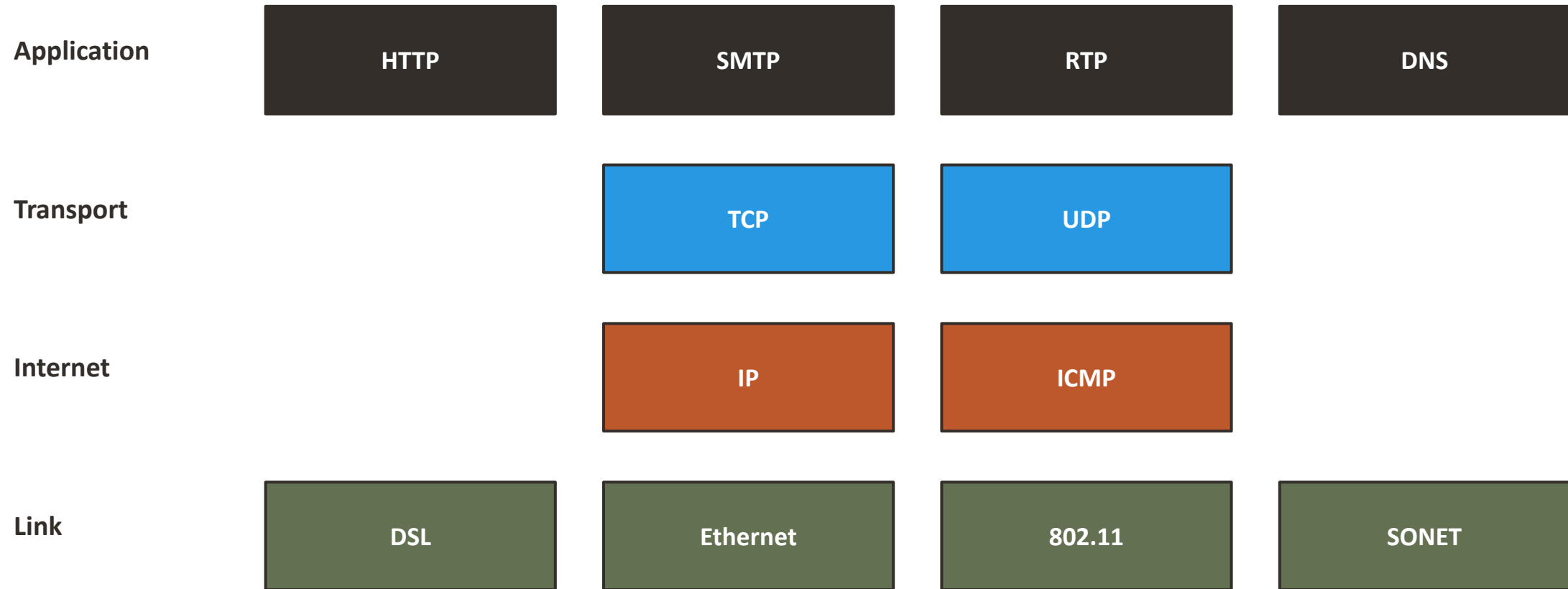
The 4-Layer Model

Internet Model, or TCP/IP Model

- The OSI model is more for educational purposes.
- The 4-layer model is what's actually used in reality.
- Application layer — corresponds to OSI's Application, Presentation, and Session layers.
- Transport layer — same as the Transport layer of OSI.
- Internet layer — same as the Network layer of OSI.
- Network/Link layer — corresponds to OSI's Physical and Data Link layers.

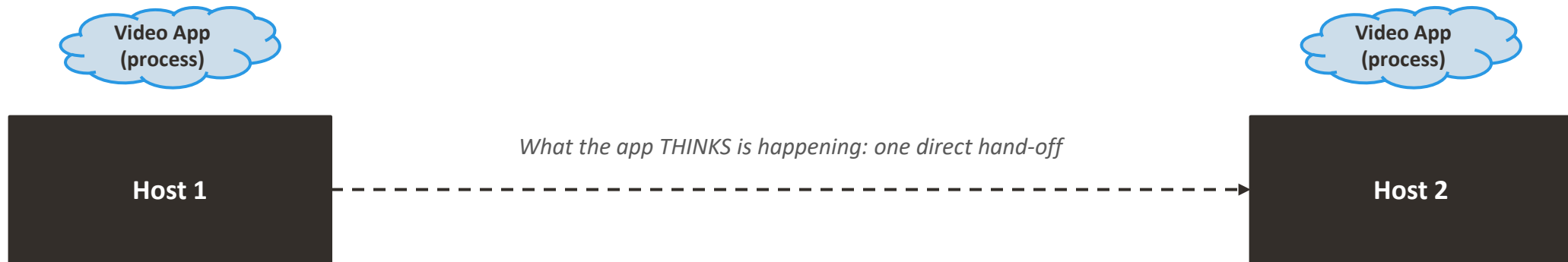


Network Protocol Stack

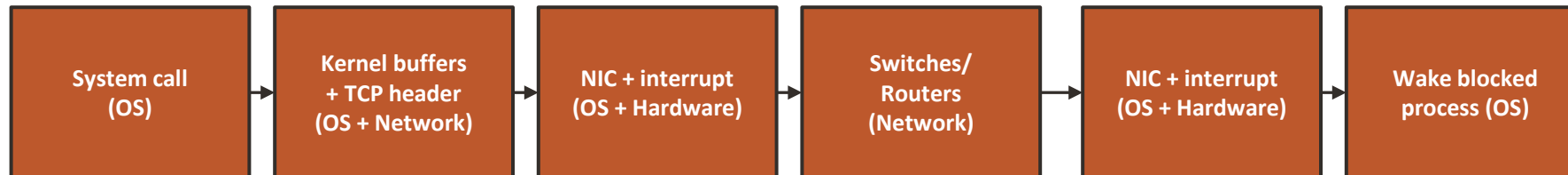


The Example: One Video-Call Frame, Two Hosts

A process on Host 1 wants to send one video frame to the same app on Host 2



What ACTUALLY happens: a relay of OS and network mechanisms, each depending on the last



Sending a Frame: The OS Side

1

The app calls send()

This is a system call — a trap into the kernel, exactly like the trap used for any I/O request.

OS: system call

2

OS builds the segment/packet

The kernel copies the frame into an OS-managed buffer, and the Transport/Network layers attach TCP and IP headers — the network protocol needs OS memory to exist in.

OS memory + L3/L4 protocol

3

OS hands data to the NIC driver

The network card is just another device the OS talks to through a driver, the same pattern used for disks.

OS: device driver

4

NIC finishes transmitting -> interrupt

The OS does not sit and wait; the process is put in the Blocked state, freeing the CPU, and a hardware interrupt later signals completion.

OS: interrupts + scheduling

Across the Network — and Back Into an OS

5

The packet crosses switches and routers

Pure networking: MAC-based forwarding at L2, IP-based routing at L3 — no single host's OS is involved here at all.

Network only

6

Host 2's NIC receives the frame -> interrupt

Exactly the same interrupt mechanism as sending, just in reverse — the OS on Host 2 must be interrupted to even notice data arrived.

OS: interrupts

7

OS decapsulates, up through the layers

The kernel strips MAC, then IP, then TCP headers, placing the payload into the socket's receive buffer — again, OS-managed memory.

OS + L2/L3/L4 protocols

8

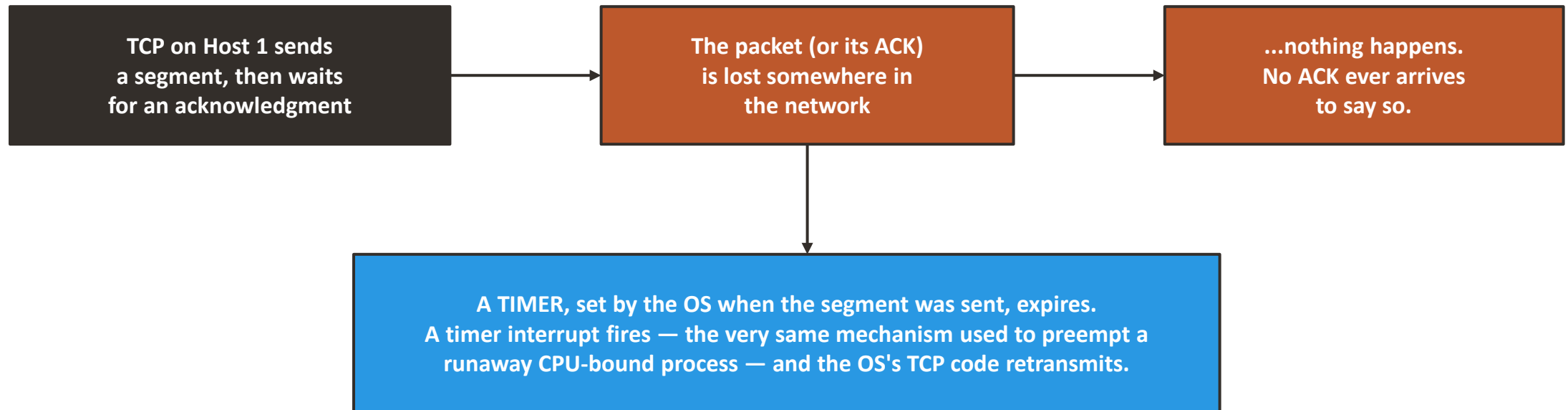
The receiving process wakes up

The app was Blocked on `recv()`; the OS moves it to Ready, and the scheduler eventually runs it, completing the illusion of one direct hand-off.

OS: process states + scheduler

What If a Packet Gets Lost?

TCP promises reliable delivery — but how does it even notice something went missing?



Key Takeaway

What the OS gives Networking

- System calls & traps to request I/O
- Interrupts to signal NIC completion / arrival
- Process states (Blocked/Ready) and the scheduler, so waiting for the network is free
- Memory management for packet buffers
- Timers, to detect and recover from packet loss

What Networking gives the OS

- A reason processes need addresses at all
- Ports — the missing piece for multiplexing many processes behind one IP
- A concrete workload that exercises interrupts, scheduling, and buffering together
- A second host to prove isolation (address spaces, protection) actually matters

Neither half of this course is self-contained: the network stack is built out of OS mechanisms, and the OS's abstractions only become truly useful once a network lets them reach beyond one machine.

Review Questions (1/2)

- Q1. A switch connects hosts H1, H2, and H3 within the network 192.168.10.x. Explain why the switch alone cannot let H2 communicate with a host on a completely different network, and name the device that makes this possible.
- Q2. Using the OSI model, explain why data sent by an application on Host 1 must pass down through Layers 7 to 1 before reaching the physical medium, and then back up through Layers 1 to 7 on Host 2, instead of going directly from Application to Application.

Review Questions (2/2)

- Q3. Describe the encapsulation process that occurs as data moves from the Application layer down to the Physical layer. In your answer, define what a Segment, a Packet, and a Frame each are, and explain how decapsulation reverses this process at the receiving host.
- Q4. Machine 234 wants to send data to Machine 345 across two routers. Explain the difference between "hop-to-hop" delivery and "end-to-end" delivery in this scenario, and identify which layer is responsible for each.
- Q5. Using the example of sending one video-call frame from Host 1 to Host 2, describe two distinct points in the journey where an OS mechanism (not a networking protocol) is essential, and explain what would go wrong at each point without it.

Thank You!